



ABM Contact Enrichment Pipeline

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1. Problem Statement

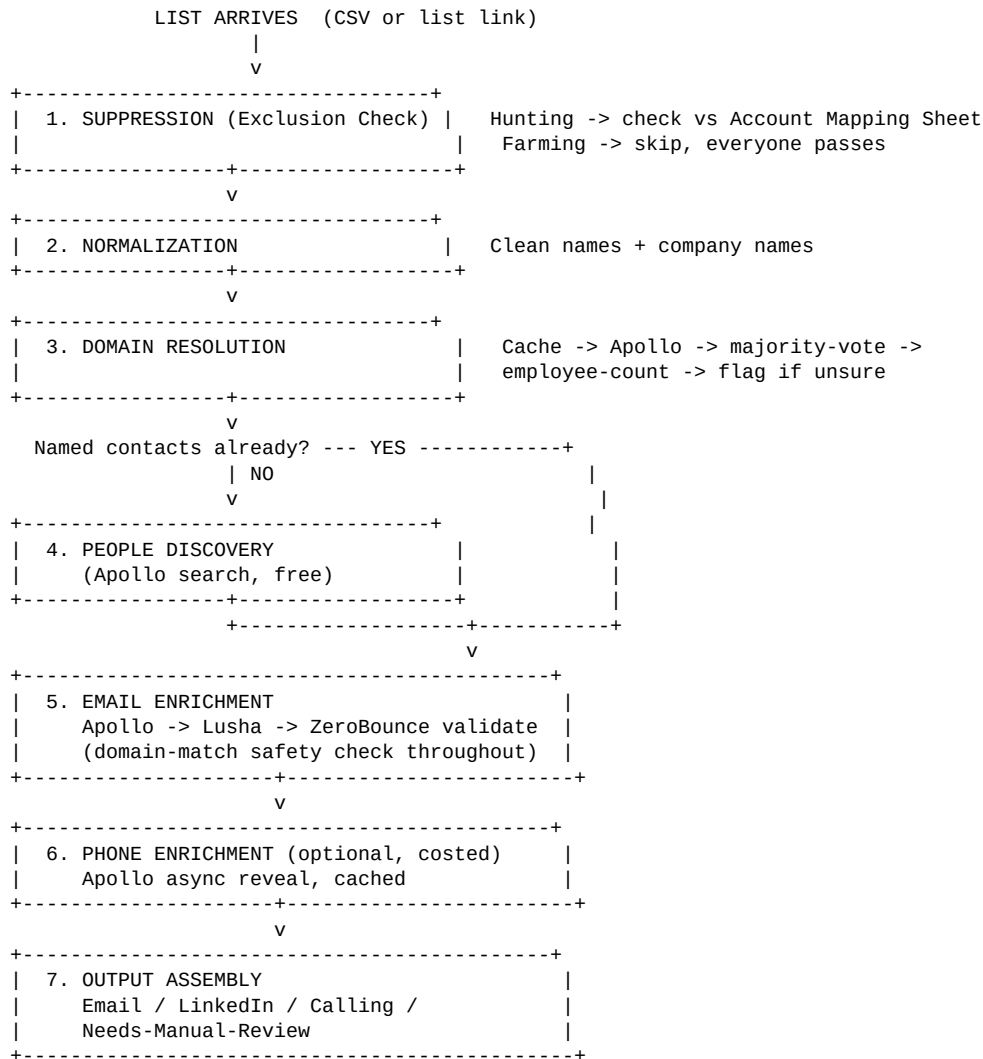
Every outbound list handed to this pipeline starts as a raw export — a conference attendee sheet, a target-account list, a company list with job titles attached. None of it is safe to load into Smartlead or HeyReach as-is: company names are unnormalized, domains are unknown or ambiguous, some fraction of the list is already a customer, and named contacts have no verified way to reach them.

Two runs this month illustrate the failure modes a manual process misses. On a 128-row UK/EUR list, 4 companies turned out to be active client relationships that would have been cold-emailed by mistake. On a 150-row Indonesia/Philippines company list, a short, generic company name (“Flip”) nearly resolved to an unrelated Indonesian fintech instead of the intended German HR-tech company — the same failure mode a human skimming a spreadsheet has no reliable way to catch.

This pipeline replaces that manual check with a sequence of stages that never guesses when it isn't confident — a wrong-company match or an unverified contact detail is treated as a bug, not an acceptable miss rate. Ambiguity gets surfaced for a five-second human decision instead of silently resolved.

2. Complete Pipeline Flow

The pipeline is a straight-line waterfall with one fork (named list vs. company-only list) and a persistent cache layer underneath every stage, so a company or contact resolved once is never re-processed or re-paid for.



3. Stage Walkthrough

Each section covers one stage — the trigger, the exact logic, and the reasoning behind the design decision.

3.1 Suppression (Exclusion Check)

Runs first, before anything else touches the list, so no downstream credits get spent on a company we shouldn't be emailing at all.

Ask which campaign type this is before running anything. **Farming** campaigns skip this stage entirely — every company passes through. **Hunting** campaigns check every company against the master Account Mapping Sheet (Client Name, Domain, Parent Company, Type 1, Parent Company Status), matching in order: exact domain → fuzzy client name (legal suffixes stripped) → fuzzy parent company. A prospect is **OK to reach out** only if the matched row's Type 1 status *and* Parent Company Status are both *Dead*. Anything else on a matched row is **Excluded** — the conservative read, since a false exclusion costs one lead, but a false inclusion means cold-emailing an existing customer. No match at all means it's a genuinely new account and passes through.

Output columns: Exclusion Status, Exclusion Reason, Matched Account.

3.2 Normalization

Cleans the surviving rows before anything gets searched against an external API — inconsistent casing and legal-entity suffixes (“Pvt Ltd”, “Inc.”, “(P) Ltd”) reduce match quality downstream, so they’re stripped here once rather than fought individually at each later stage.

Splits a Full Name into First/Last (first token → first name, everything else → last name). Strips honorific prefixes and degree/generational suffixes from names. Strips legal-entity suffixes from company names. Originals are always preserved alongside the cleaned columns, never overwritten.

3.3 Domain Resolution

The highest-risk stage in the pipeline — a wrong domain here means every downstream email search targets the wrong real company. Never guesses when uncertain.

Checks the shared cache first (instant, free, and grows with every list run). On a cache miss, searches Apollo's organization directory by the cleaned company name. If Apollo returns exactly one exact-name match, accept it. If it returns several, first check whether they already agree on the same domain — large companies frequently have duplicate/fragmented org records under an identical display name, and treating that as ambiguous would falsely block obvious cases (Ernst & Young showed up as 18 near-duplicate records, all pointing to ey . com). Only when the candidates genuinely disagree on domain does it fall to employee-count comparison against the source list's headcount column, and only when *that* produces a clear winner does it accept automatically. Anything left — no employee-count signal, or a real split decision — is flagged **Ambiguous** with the full candidate list, for a human to pick in seconds rather than the pipeline guessing wrong. Zero exact-name matches at all is flagged **Unresolved**, which typically means a small/regional company or a stale legacy brand name that needs a quick web search.

Output columns: Resolved Domain, Resolved Company LinkedIn, Resolution Source.

3.4 People Discovery

Only runs when the input list is company names without named contacts. Runs before any credits are spent — a person-search query against Apollo is free; only revealing a specific person's contact details costs credits.

Searches Apollo by company domain plus the target job titles, with a seniority filter (c_suite, vp, head, director, manager) applied at the query level — without it, a single page of title-matched results at a large company gets swamped by hundreds of lower-level hits and the actual decision-maker never surfaces. Search results come back with an obfuscated last name (e.g. “Laura Bu***r”) plus flags for whether Apollo has email/phone data on file — real names and contact details are revealed in a separate, credit-costed step. Candidates are ranked by how senior their title is, one distinct seniority tier covered before doubling up on any tier, up to a configurable cap per company.

3.5 Email Enrichment

The core enrichment step, and the one that most needs its own safety net — a “verified” email from any single provider is not proof it belongs to the company you're searching for.

Calls Apollo's contact match (by Apollo ID when available, otherwise name + domain) and checks the returned email's domain against the company domain being searched. A match is accepted outright; a known alias (a confirmed rebrand, an M&A absorption, or a company that genuinely emails from a different domain than its marketing site) is also accepted; a same-brand name under a different country code (any large multinational professional-services network) is also accepted. Anything else is held as a **mismatch** for manual review rather than trusted or discarded automatically. Contacts Apollo can't find an email for at all fall to a second waterfall step through Lusha, using the same domain-match check, and any Lusha hit must additionally pass ZeroBounce validation before it's accepted.

3.6 Phone Enrichment

Optional, and meaningfully more expensive than email — treat this as a deliberate scope decision, not something to default on for a large list.

Apollo's phone reveal is asynchronous: it never returns a number in the direct response, only via a webhook callback seconds later, so this stage needs an internet-facing relay to receive it. Confirmed through testing: Apollo does not call the webhook at all when it has no phone data — a missing callback means “nothing on file,” not a slow response, so the wait window can stay short. Every successful reveal is cached (keyed by contact, not by list) so the same person is never paid for twice across runs, even in a future, unrelated list.

3.7 Output Assembly

Splits the fully-enriched list into channel-specific files so each downstream tool only sees what it needs, plus a bucket for anything that didn't clear the bar. Nothing that fails a check gets silently dropped — every non-accepted row lands in the Needs Manual Review file with the reason attached.

4. Script Specification

Stage	Script	Calls	Output
Suppression	abm-exclusion-check skill	— (local match)	Exclusion Status / Reason / Matched Account
Normalization	name-company-normalizer skill	— (local match)	Cleaned First/Last/Company Name
Domain Resolution	resolve_company_domains.py	Apollo Org Search	Resolved Domain / LinkedIn / Source
People Discovery	search_company_contacts_apollo.py	Apollo People Search (free)	apollo_id / title / persona_tier / has_email
Email Enrichment	enrich_contacts_apollo.py	Apollo People Match	email / email_status / linkedin_apollo / note
Email Waterfall	lusha_waterfall_enrich.py	Lusha Search & Enrich + ZeroBounce	lusha_email / zerobounce_status / lusha_note
Phone Enrichment	enrich_phone_apollo.py	Apollo Match (async) + webhook.site	Phone Number / Type / Confidence / Note
Phone Comparison	lusha_phone_compare.py	Lusha Search & Enrich	Apollo vs. Lusha phone side-by-side

5. Configuration Reference

Every knob worth tuning, where it lives, and what changing it actually does.

Stage	Parameter	Location	Default	Controls
Suppression	Campaign type	Asked at runtime	—	Whether exclusion logic applies at all (Farming = skip)
Suppression	Fuzzy match threshold	check_exclusions.py	0.88	How lenient company-name matching is
Domain Resolution	Shared cache	reference/company_domain_cache.csv	grows over time	Instant, free resolution for any company seen before
Domain Resolution	Majority-domain threshold	resolve_company_domain_mains.py	>50% of candidates	Duplicate records treated as agreement vs. real ambiguity
Domain Resolution	Employee-count ambiguity gate	resolve_company_domain_mains.py	gap>0.5, diff<0.15	When to flag instead of auto-picking by headcount
Domain Resolution	Legal-suffix list	resolve_company_domain_mains.py	Inc/Ltd/Corp/PLC/etc.	Which suffixes get stripped before name comparison
Domain Resolution	Apollo search page size	resolve_company_domain_mains.py	25	How many candidates get pulled before giving up
People Discovery	Target job titles	PERSONAS list	17 HR-function titles	Which roles get searched for at each company
People Discovery	Seniority filter	search_company_contacts_apollo.py	c_suite, vp, head, director, manager	Which seniority levels Apollo returns
People Discovery	Max contacts per company	MAX_PER_COMPANY	10	Contact volume cap per company
Email Enrichment	Known domain aliases	reference/domain_aliases.csv	grows over time	Which mismatched domains are actually legitimate
Email Enrichment	Country-variant matching	domain_root() function	root name ≥4 chars	Auto-accepts same-brand, different-country-TLD emails
Email Waterfall	Run Lusha at all?	Asked at runtime	—	Whether to spend Lusha credits on Apollo misses
Email Waterfall	ZeroBounce acceptance	lusha_waterfall_enrich.py	valid, catch-all	How strict validation is before accepting a Lusha email
Phone Enrichment	Run phone at all?	Asked at runtime	off by default	Cost gate — phone is ~8x the credit cost of email
Phone Enrichment	Poll timeout	POLL_TIMEOUT	30s	How long to wait for Apollo's async webhook callback
Phone Enrichment	Persistent cache	reference/phone_reveal_cache.csv	grows over time	Never re-pay for a phone number already revealed
Phone Enrichment	Cache invalidation	Manual — delete the row	none automatic	No job-change detection; stale numbers cleared by hand

6. Technical Implementation

Layer	Responsibility	Technology
Orchestrator	Decide campaign type, sequence stages, gate costly steps behind explicit go-ahead	Agent session (this pipeline)
Suppression	Match prospects against the master Account Mapping Sheet	Python + fuzzy match
Normalization	Clean names and company names	Python
Domain Resolution	Resolve and cache company domains	Apollo Org Search API + persistent CSV cache
People Discovery	Find named contacts at target companies	Apollo People Search API (no cost)
Email Enrichment	Reveal and validate verified emails	Apollo People Match → Lusha → ZeroBounce
Phone Enrichment	Reveal verified mobile numbers	Apollo async reveal via webhook relay + cache
Output	Assemble channel-specific deliverables	Python / pandas

6.1 Enrichment Waterfall Pipeline

Step	Action	Details
1. SUPPRESS	Check against client list	Domain / name / parent-company match, Hunting campaigns only
2. CLEAN	Normalize names and companies	Strip suffixes, split full names
3. RESOLVE	Find each company's domain	Cache → Apollo → majority-vote → employee-count → flag
4. DISCOVER	Find people, if not already named	Apollo search by title + seniority, free
5. VERIFY EMAIL	Reveal and validate	Apollo → Lusha → ZeroBounce
6. VERIFY PHONE	Reveal, optional	Apollo webhook reveal, cached per contact
7. ASSEMBLE	Build deliverables	Email / LinkedIn / Calling / Needs Manual Review

6.2 Environment & API Requirements

Requirement	Details
API keys	APOLLO_API_KEY, LUSHA_API_KEY, ZEROBOUNCE_API_KEY, SMARTLEAD_API_KEY, HEYREACH_API_KEY — all in .env, never printed
Reference files	company_domain_cache.csv, phone_reveal_cache.csv, domain_aliases.csv — self-updating, shared across future lists
Master data source	Account Mapping Sheet, sourced from HubSpot (read-only — never written back to)
Cost model	Apollo email ≈1 credit/hit · Apollo phone ≈8 credits/hit · Lusha ≈6 credits/hit · ZeroBounce ≈1/check

Requirement	Details
Third-party relay	webhook.site — created fresh per phone-enrichment run, deleted immediately after; no permanent PII storage